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| 5 | <p>(a) When a wave (front) is incident on an edge or an obstacle/slit/gap</p> <p>Wave 'bends' into the geometrical shadow/changes direction/spreads</p> | <p>M1</p> <p>A1</p> | <p>[2]</p> |
| | <p>(b) (i) $d = 1/(750 \times 10^3)$
 $= 1.33 \times 10^{-6} \text{ m}$</p> | <p>C1</p> <p>A1</p> | <p>[2]</p> |
| | <p>(ii) $1.33 \times 10^{-6} \times \sin 90^\circ = n \times 590 \times 10^{-9}$
 $n = 2$ (must be an integer)</p> | <p>C1</p> <p>A1</p> | <p>[2]</p> |
| | <p>(iii) formula assumes no path difference of light before entering grating <u>or</u>
 there is a path difference before the grating</p> | <p>B1</p> | <p>[1]</p> |
| | <p>(c) e.g. lines further apart in second order
 lines fainter in second order
 (allow any sensible difference: 1 each, max 2)
 (if differences stated but without reference to the orders, max 1 mark)</p> | <p>B2</p> | <p>[2]</p> |